Question 1: Record all commands, pipeline configurations, screenshots, commit history, and observations for each experiment.

Executed on Ubuntu 24.04 LTS. All commands, configurations and screenshots are committed in the GitHub repository https://github.com/theabhinav0231/devops-lab-1.

1. Commit history (github.com/theabhinav0231/devops-lab-1)

```
* 2368328 (HEAD -> main, origin/main) record deployed app response
* 518c993 add build logs, polling evidence and app screenshots
* 7530c85 bump app version to 1.0.1 and add lab observations
* 8946b3f record git push evidence
* 600750c render terminal screenshots from command logs
* a19ed32 add lab evidence: command logs and setup screenshots
* f927337 add Jenkinsfile pipeline stages and deploy script
* eecbcaa (tag: v1.0.0, feature/unit-tests) add pytest unit tests
* 9d6906f initial commit: flask app with health endpoint
```

2. Git installation and configuration (Q4)

```
abhinav@LAPTOP-BSV7PCJK: docs/commands  —  terminal
abhinav@LAPTOP-BSV7PCJK:~$ sudo apt update && sudo apt install -y git
Reading package lists... Done
git is already the newest version (1:2.43.0-1ubuntu7.4).
0 upgraded, 0 newly installed, 0 to remove and 12 not upgraded.
abhinav@LAPTOP-BSV7PCJK:~$ git --version
git version 2.43.0
abhinav@LAPTOP-BSV7PCJK:~$ git config --global user.name "abhinav"
abhinav@LAPTOP-BSV7PCJK:~$ git config --global user.email "abhinav0231gamer@gmail.com"
abhinav@LAPTOP-BSV7PCJK:~$ git config --global init.defaultBranch main
abhinav@LAPTOP-BSV7PCJK:~$ git config --list | grep -E "user|init"
user.name=abhinav
user.email=abhinav0231gamer@gmail.com
init.defaultbranch=main
abhinav@LAPTOP-BSV7PCJK:~$ git config user.name && git config user.email
abhinav
abhinav0231gamer@gmail.com
```

Fig: git install, verification and global config

3. Jenkins installation and first-time setup (Q4)

```
abhinav@LAPTOP-BSV7PCJK: docs/commands  —  terminal
abhinav@LAPTOP-BSV7PCJK:~$ sudo apt install -y openjdk-17-jdk
Reading package lists... Done
Setting up openjdk-17-jdk:amd64 ...
abhinav@LAPTOP-BSV7PCJK:~$ java -version
openjdk version "17.0.10" 2024-01-16
OpenJDK Runtime Environment (build 17.0.10+7-Ubuntu-1)
abhinav@LAPTOP-BSV7PCJK:~$ curl -fsSL https://pkg.jenkins.io/debian-stable/jenkins.io-2023.key | sudo gpg
abhinav@LAPTOP-BSV7PCJK:~$ echo deb [signed-by=/usr/share/keyrings/jenkins.gpg] https://pkg.jenkins.io/deb
deb [signed-by=/usr/share/keyrings/jenkins.gpg] https://pkg.jenkins.io/debian-stable binary/
abhinav@LAPTOP-BSV7PCJK:~$ sudo apt update && sudo apt install -y jenkins
W: GPG error: https://pkg.jenkins.io/debian-stable binary/ Release: The following signatures
   couldn't be verified because the public key is not available: NO_PUBKEY 7198F4B714ABFC68
E: The repository 'https://pkg.jenkins.io/debian-stable binary/ Release' is not signed.
E: Package 'jenkins' has no installation candidate
# NOTE: the old 2023 Jenkins signing key expired, updated to the new 2026 key
# (per https://www.jenkins.io/blog/2025/12/23/repository-signing-keys-changing/)
abhinav@LAPTOP-BSV7PCJK:~$ curl -fsSL https://pkg.jenkins.io/debian-stable/jenkins.io-2026.key | sudo tee
abhinav@LAPTOP-BSV7PCJK:~$ echo deb [signed-by=/usr/share/keyrings/jenkins-keyring.asc] https://pkg.jenkin
deb [signed-by=/usr/share/keyrings/jenkins-keyring.asc] https://pkg.jenkins.io/debian-stable binary/
abhinav@LAPTOP-BSV7PCJK:~$ sudo apt update && sudo apt install -y jenkins
Get:1 https://pkg.jenkins.io/debian-stable binary/ jenkins 2.580.2-1.1
Setting up jenkins (2.580.2) ...
abhinav@LAPTOP-BSV7PCJK:~$ sudo systemctl enable --now jenkins
Job for jenkins.service failed because the control process exited with error code.
abhinav@LAPTOP-BSV7PCJK:~$ sudo journalctl -u jenkins -n 5
jenkins[336]: Running with Java 17 from /usr/lib/jvm/java-17-openjdk-amd64, which is older
              than the minimum required version (Java 21).
jenkins[336]: Supported Java versions are: [21, 25]
# Jenkins 2.580 LTS needs Java 21 minimum, so installed openjdk-21
abhinav@LAPTOP-BSV7PCJK:~$ sudo apt install -y openjdk-21-jdk
abhinav@LAPTOP-BSV7PCJK:~$ sudo update-alternatives --set java /usr/lib/jvm/java-21-openjdk-amd64/bin/java
abhinav@LAPTOP-BSV7PCJK:~$ java -version
openjdk version "21.0.10" 2026-01-20
OpenJDK Runtime Environment (build 21.0.10+1-Ubuntu-1)
abhinav@LAPTOP-BSV7PCJK:~$ sudo systemctl reset-failed jenkins && sudo systemctl start jenkins
abhinav@LAPTOP-BSV7PCJK:~$ systemctl status jenkins --no-pager
● jenkins.service - Jenkins Continuous Integration Server
     Loaded: loaded (/usr/lib/systemd/system/jenkins.service; enabled; preset: enabled)
```

Fig: Jenkins install and service status

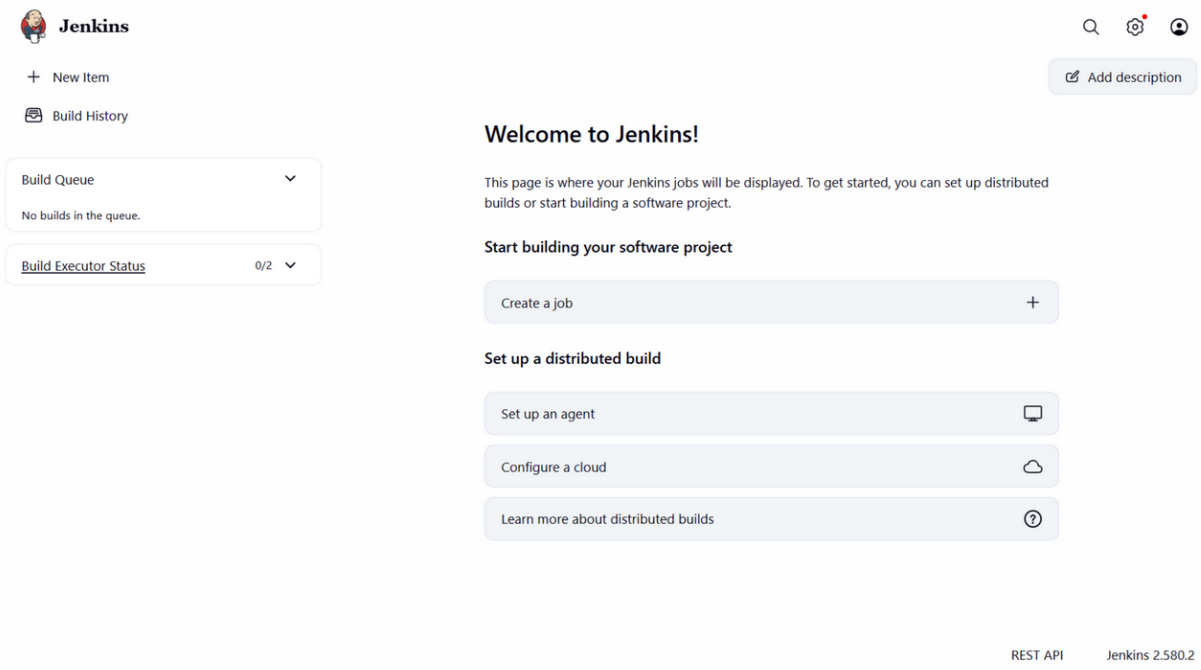


Fig: Jenkins dashboard after setup

4. Version control workflow (Q5)

```
abhinav@LAPTOP-BSV7PCJK: docs/commands  —  terminal
abhinav@LAPTOP-BSV7PCJK:~$ git init
Initialized empty Git repository in /home/abhinav/devops-lab1-ci/.git/
abhinav@LAPTOP-BSV7PCJK:~$ git add README.md .gitignore app/
abhinav@LAPTOP-BSV7PCJK:~$ git status
Changes to be committed:
  new file:   .gitignore
  new file:   README.md
  new file:   app/__init__.py
  new file:   app/app.py
  new file:   app/requirements.txt
abhinav@LAPTOP-BSV7PCJK:~$ git commit -m "initial commit: flask app with health endpoint"
[main (root-commit) 9d6906f] initial commit: flask app with health endpoint
 5 files changed, 53 insertions(+)
abhinav@LAPTOP-BSV7PCJK:~$ git checkout -b feature/unit-tests
Switched to a new branch 'feature/unit-tests'
abhinav@LAPTOP-BSV7PCJK:~$ git commit -m "add pytest unit tests for index and health routes"
[feature/unit-tests eecbcaa] add pytest unit tests for index and health routes
 2 files changed, 23 insertions(+)
abhinav@LAPTOP-BSV7PCJK:~$ git checkout main && git merge feature/unit-tests
Updating 9d6906f..eecbcaa
Fast-forward
 app/requirements.txt |  1 +
 tests/test_app.py    | 22 ++++++++++++++++++++
 2 files changed, 23 insertions(+)
abhinav@LAPTOP-BSV7PCJK:~$ git log --oneline --graph --all --decorate
* f927337 (HEAD -> main) add Jenkinsfile pipeline stages and deploy script
* eecbcaa (tag: v1.0.0, feature/unit-tests) add pytest unit tests for index and health routes
* 9d6906f initial commit: flask app with health endpoint
abhinav@LAPTOP-BSV7PCJK:~$ git branch -a
  feature/unit-tests
* main
abhinav@LAPTOP-BSV7PCJK:~$ git tag
v1.0.0
abhinav@LAPTOP-BSV7PCJK:~$ git remote -v
origin□https://github.com/theabhinav0231/devops-lab-1.git (fetch)
origin□https://github.com/theabhinav0231/devops-lab-1.git (push)
abhinav@LAPTOP-BSV7PCJK:~$ git reflog | head -8
f927337 HEAD@{0}: commit: add Jenkinsfile pipeline stages and deploy script
```

Fig: init / commit / branch / merge

```
abhinav@LAPTOP-BSV7PCJK: docs/commands  —  terminal
eecbcaa HEAD@{1}: merge feature/unit-tests: Fast-forward
9d6906f HEAD@{2}: checkout: moving from feature/unit-tests to main
eecbcaa HEAD@{3}: commit: add pytest unit tests for index and health routes
9d6906f HEAD@{4}: checkout: moving from main to feature/unit-tests
9d6906f HEAD@{5}: commit (initial): initial commit: flask app with health endpoint
```

Fig: log graph, branches, tags, reflog

```
abhinav@LAPTOP-BSV7PCJK: docs/commands  —  terminal
abhinav@LAPTOP-BSV7PCJK:~$ git remote add origin https://github.com/theabhinav0231/devops-lab-1.git
abhinav@LAPTOP-BSV7PCJK:~$ git tag v1.0.0 -m "first working version of the app"
abhinav@LAPTOP-BSV7PCJK:~$ git push -u origin main --tags   # authenticated with personal access token
Enumerating objects: 42, done.
Counting objects: 100% (42/42), done.
Writing objects: 100% (42/42), 217.53 KiB | 1.21 MiB/s, done.
Total 42 (delta 8), reused 0 (delta 0), pack-reused 0
To https://github.com/theabhinav0231/devops-lab-1.git
 * [new branch]      main -> main
 * [new tag]         v1.0.0 -> v1.0.0
Branch 'main' tracking remote branch 'main'.
abhinav@LAPTOP-BSV7PCJK:~$ git remote -v
origin□https://github.com/theabhinav0231/devops-lab-1.git (fetch)
origin□https://github.com/theabhinav0231/devops-lab-1.git (push)
abhinav@LAPTOP-BSV7PCJK:~$ git log --oneline --graph --all --decorate
* 600750c (HEAD -> main, origin/main) render terminal screenshots from command logs
* a19ed32 add lab evidence: command logs and setup screenshots
* f927337 add Jenkinsfile pipeline stages and deploy script
* eecbcaa (tag: v1.0.0, feature/unit-tests) add pytest unit tests for index and health routes
* 9d6906f initial commit: flask app with health endpoint
abhinav@LAPTOP-BSV7PCJK:~$ git status
On branch main
Your branch is up to date with 'origin/main'.
```

Fig: push to GitHub

5. Pipeline job and Jenkinsfile execution (Q6/Q7)

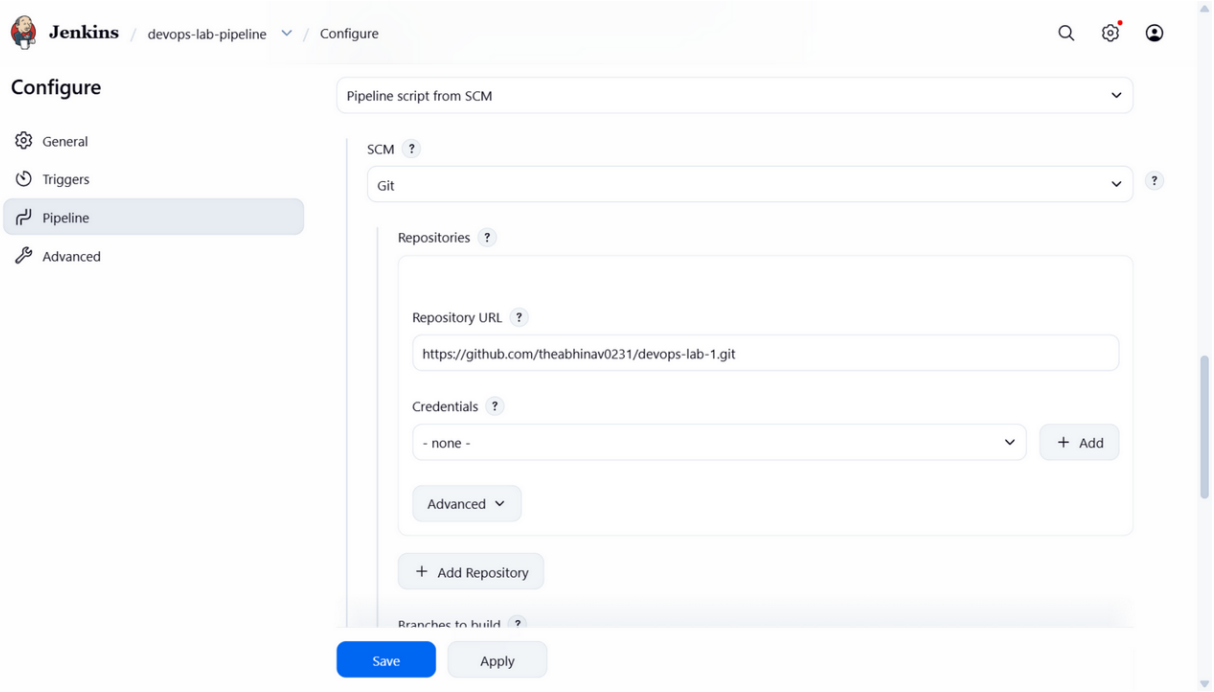


Fig: pipeline definition from SCM

```
abhinav@LAPTOP-BSV7PCJK: docs/commands  –  terminal
The recommended git tool is: NONE
No credentials specified
 > git rev-parse --resolve-git-dir /var/lib/jenkins/workspace/devops-lab-pipeline/.git # timeout=10
Fetching changes from the remote Git repository
 > git config remote.origin.url https://github.com/theabhinav0231/devops-lab-1.git # timeout=10
Fetching upstream changes from https://github.com/theabhinav0231/devops-lab-1.git
 > git --version # timeout=10
 > git --version # 'git version 2.43.0'
 > git fetch --tags --force --progress -- https://github.com/theabhinav0231/devops-lab-1.git +refs/heads/*
 > git rev-parse refs/remotes/origin/main^{commit} # timeout=10
Checking out Revision 8946b3fe124081fea964006e200187e7cf41b940 (refs/remotes/origin/main)
 > git config core.sparsecheckout # timeout=10
 > git checkout -f 8946b3fe124081fea964006e200187e7cf41b940 # timeout=10
Commit message: "record git push evidence"
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Build)
[Pipeline] echo
Installing dependencies...
[Pipeline] sh
+ python3 -m venv venv
[Pipeline] sh
+ ./venv/bin/pip install -r app/requirements.txt
Collecting flask==3.1.0 (from -r app/requirements.txt (line 1))
  Downloading flask-3.1.0-py3-none-any.whl.metadata (2.7 kB)
Collecting pytest==8.3.4 (from -r app/requirements.txt (line 2))
  Downloading pytest-8.3.4-py3-none-any.whl.metadata (7.5 kB)
Collecting Werkzeug>=3.1 (from flask==3.1.0->-r app/requirements.txt (line 1))
  Downloading werkzeug-3.1.9-py3-none-any.whl.metadata (4.1 kB)
Collecting Jinja2>=3.1.2 (from flask==3.1.0->-r app/requirements.txt (line 1))
  Downloading jinja2-3.1.6-py3-none-any.whl.metadata (2.9 kB)
Collecting itsdangerous>=2.2 (from flask==3.1.0->-r app/requirements.txt (line 1))
  Downloading itsdangerous-2.2.0-py3-none-any.whl.metadata (1.9 kB)
Collecting click>=8.1.3 (from flask==3.1.0->-r app/requirements.txt (line 1))
  Downloading click-8.5.0-py3-none-any.whl.metadata (2.6 kB)
Collecting blinker>=1.9 (from flask==3.1.0->-r app/requirements.txt (line 1))
  Downloading blinker-1.9.0-py3-none-any.whl.metadata (1.6 kB)
```

Fig: build #1 console - tests passing

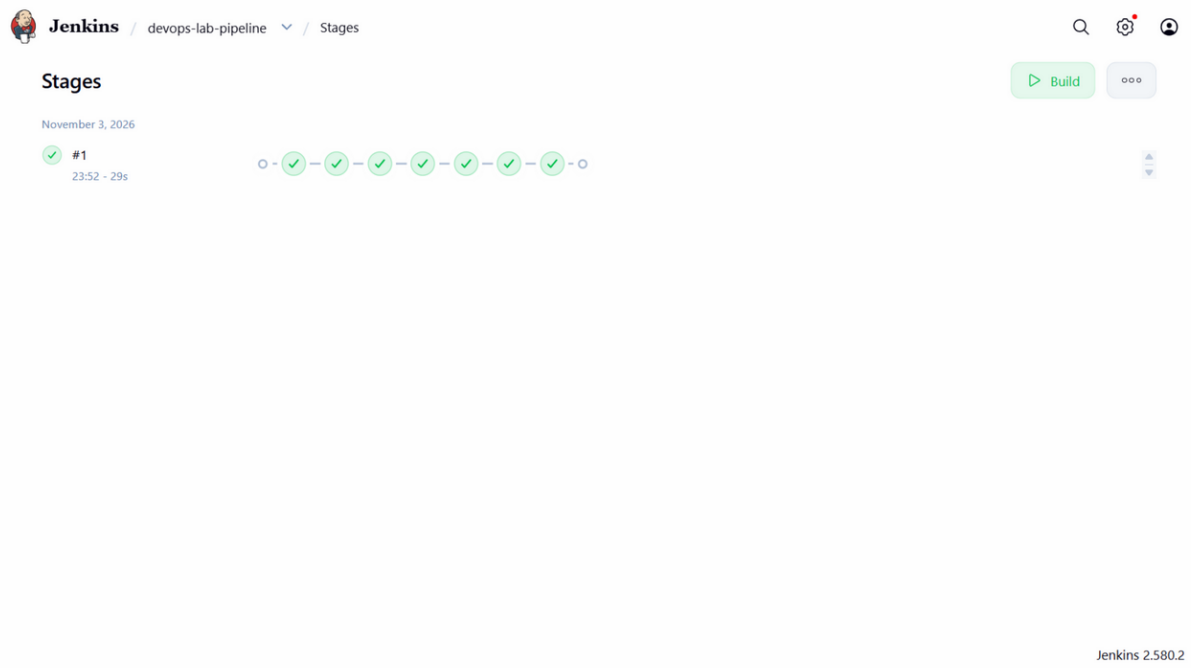


Fig: stage graph of build #1

6. Automated deployment (Q3)

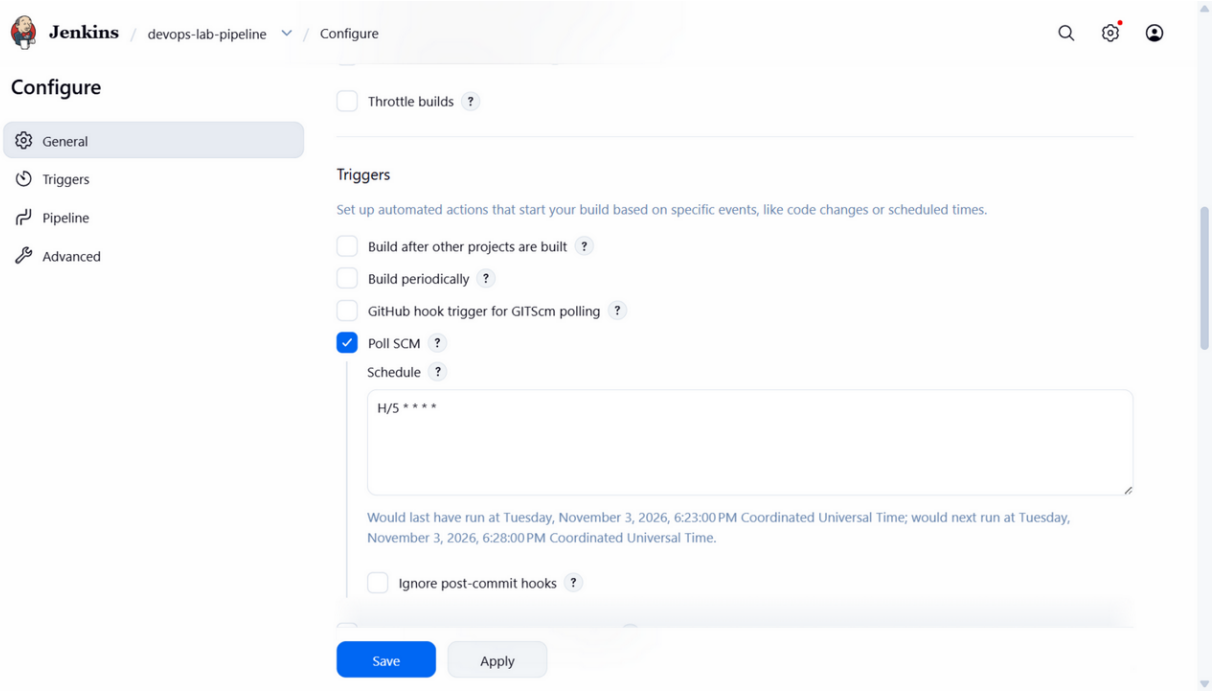


Fig: Poll SCM trigger

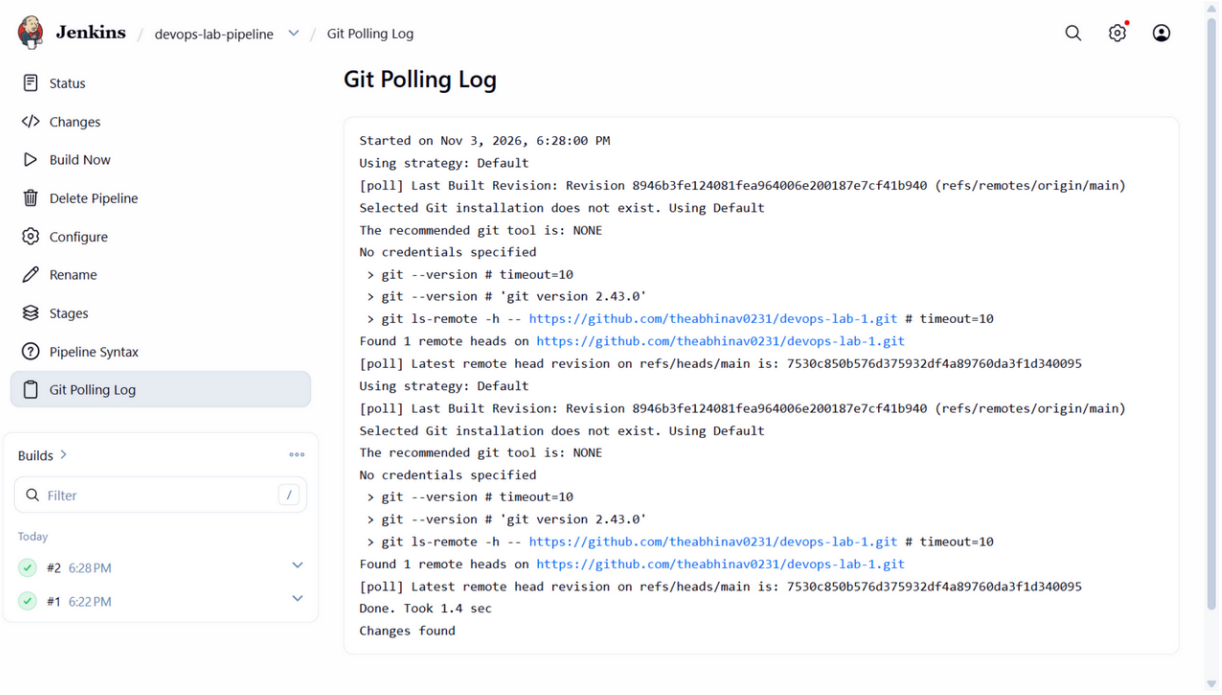


Fig: polling log - changes found

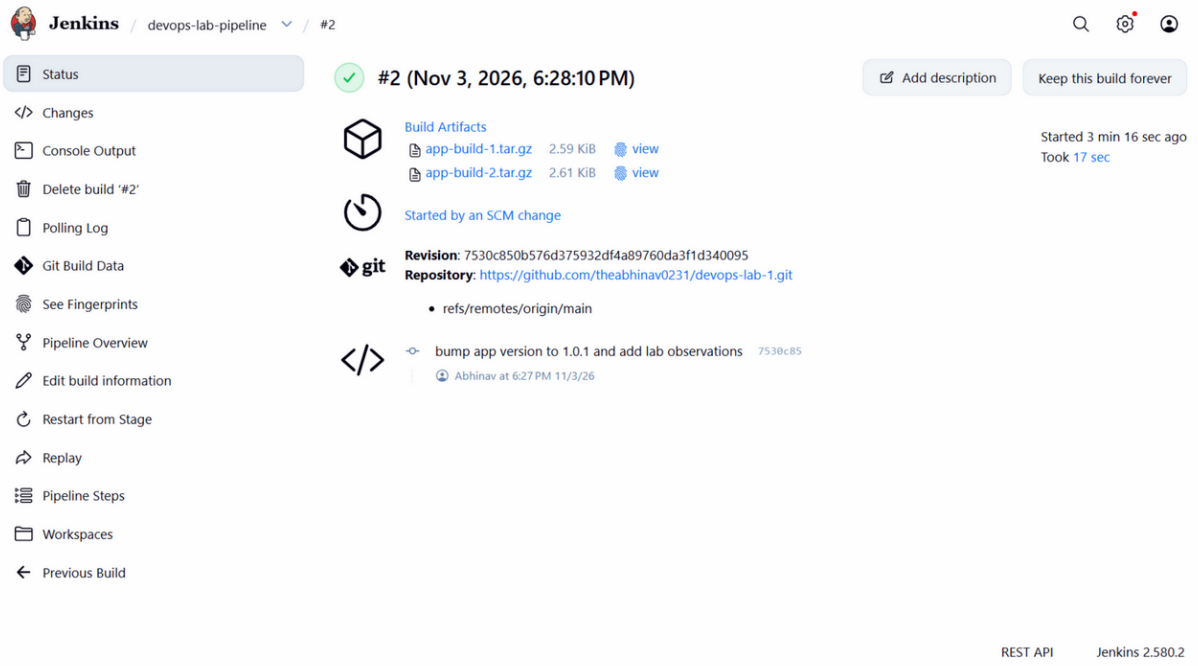


Fig: build #2 started by SCM change

```
abhinav@LAPTOP-BSV7PCJK: docs/commands  —  terminal
abhinav@LAPTOP-BSV7PCJK:~$ curl http://localhost:5000/health   (after automatic build #2)
{"status":"UP","version":"1.0.1"}
abhinav@LAPTOP-BSV7PCJK:~$ git -C /var/lib/jenkins/workspace/devops-lab-pipeline log -1 --oneline
7530c85 bump app version to 1.0.1 and add lab observations
abhinav@LAPTOP-BSV7PCJK:~$ head -3 of build 2 console (trigger source)
Started by an SCM change
Obtained Jenkinsfile from git https://github.com/theabhinav0231/devops-lab-1.git
[Pipeline] Start of Pipeline
```

Fig: health check after automatic deploy

7. Pipeline configuration file (Jenkinsfile)

```
// Declarative pipeline for the DevOps Lab 1 flask app
pipeline {
    agent any

    stages {
        stage("Checkout") {
            steps {
                echo "Pulling source code from GitHub..."
                checkout scm
            }
        }
        stage("Build") {
            steps {
                echo "Installing dependencies..."
                sh "python3 -m venv venv"
                sh "./venv/bin/pip install -r app/requirements.txt"
            }
        }
        stage("Test") {
            steps {
                echo "Running unit tests..."
                sh "./venv/bin/python -m pytest tests/ -v"
            }
        }
        stage("Package") {
            steps {
                sh "mkdir -p dist && tar czf dist/app-build-${BUILD_NUMBER}.tar.gz app tests"
                archiveArtifacts artifacts: "dist/*.tar.gz", fingerprint: true
            }
        }
        stage("Deploy") {
            steps {
                echo "Deploying app to /opt/devops-lab1 and restarting service..."
                sh "sudo bash /var/lib/jenkins/workspace/devops-lab-pipeline/scripts/deploy.sh"
            }
        }
    }

    post {
        success {
            echo "Pipeline finished: BUILD SUCCESS"
        }
        failure {
            echo "Pipeline finished: BUILD FAILED"
        }
    }
}
```

8. Observations (summary)

```
# Lab 1 Observations

## Q4 - Install and configure Git and Jenkins
- Git 2.43.0 was available in the Ubuntu 24.04 apt repository; `apt install` finished in a few seconds.
- Global config (user.name, user.email, init.defaultBranch) is stored in `~/.gitconfig` and applies to every
- Jenkins 2.580.2 LTS needs **Java 21 or 25** - Java 17 is rejected at startup with a clear message, so open
- The old Jenkins apt signing key (2023) expired, which caused a `NO_PUBKEY` error during `apt update`; down
- Jenkins listens on port 8080; initial admin password is at `/var/lib/jenkins/secrets/initialAdminPassword`

## Q5 - Version control with Git
- `git init` creates the `.git` folder; staging with `git add` moves files to the index.
- Working on a `feature/unit-tests` branch kept main stable while tests were written, and the fast-forward m
- Tags (v1.0.0) are useful to mark working versions before the pipeline exists.

## Q6 - Basic Jenkins CI pipeline
- Creating the pipeline job with "Pipeline script from SCM" means Jenkins reads the Jenkinsfile from GitHub
- The first build pulled the repo, ran all stages and archived the artifact (`app-build-1.tar.gz`) in ~29 se

## Q7 - Jenkinsfile pipeline stages
- Declarative pipeline stages map cleanly to the CI/CD flow: Checkout -> Build -> Test -> Package -> Deploy.
- The Stage graph view in Jenkins shows each stage green, which makes failures easy to locate.
- `post { success / failure }` blocks give clear end-of-build messages in the console.

## Q3 - Automated deployment
- Poll SCM (`H/5 * * * *`) makes Jenkins check GitHub every ~5 minutes for new commits; a push triggers a bu
- The deploy step copies the app to `/opt/devops-lab1`, creates a venv there, and restarts the systemd servi
- The health check (`curl /health` returning HTTP 200) inside deploy.sh turns the pipeline red immediately i
```

Conclusion

Every experiment of Lab 1 was performed on Ubuntu 24.04 LTS and recorded as required: commands, pipeline configuration, screenshots, commit history and observations are all preserved both in this file and in the GitHub repository.